

ULLARED, Sweden

WMO: 025390

Lat: 57.1134N

Lon: 12.7732E

Elev: 400

StdP: 14.48

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.5	10.9	2.3	6.3	6.8	7.0	8.0	14.7	12.2	34.0	11.2	34.2	0.6	40	0.540

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.2	80.0	64.5	76.7	63.4	73.2	61.9	67.2	75.4	65.5	73.1	63.7	70.5	3.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	92.1	70.0	62.7	86.5	68.0	61.1	81.5	66.5	31.9	75.5	30.5	73.3	29.2	70.8	75.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.9	8.8	7.6	DB	0.1	84.0	6.6	3.8	-4.6	86.8	-8.4	89.0	-12.1	91.1	-16.9	93.9	(4)
			WB	-0.9	71.0	6.5	2.2	-5.5	72.6	-9.3	73.9	-13.0	75.1	-17.7	76.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.4	28.4	29.9	34.8	42.5	50.7	56.5	60.8	59.5	53.0	44.8	38.2	32.9
	DBStd	12.93	8.30	7.25	6.77	6.06	6.94	5.29	4.94	4.53	5.04	6.56	6.81	8.74
	HDD50	3122	669	563	472	236	74	6	0	1	29	186	356	530
	HDD65	7554	1134	983	937	675	447	260	151	179	361	627	805	995
	CDD50	1080	0	0	0	11	95	201	335	296	118	23	1	0
	CDD65	37	0	0	0	0	3	5	21	8	0	0	0	0
	CDH74	606	0	0	0	1	53	148	309	90	5	0	0	0
	CDH80	89	0	0	0	0	3	18	60	8	0	0	0	0
Wind	WSAvg	3.2	3.6	3.7	3.5	3.5	3.1	2.6	2.3	2.4	2.7	3.1	3.7	3.7
Precipitation	PrecAvg	45.6	4.3	3.2	2.8	2.4	2.8	3.7	4.3	4.6	4.3	4.8	4.1	4.3
	PrecMax	57.3	8.5	6.3	6.9	5.1	4.4	6.4	9.6	7.8	10.2	10.2	7.0	9.8
	PrecMin	32.1	0.1	0.9	1.0	0.4	0.5	0.0	0.8	0.7	1.0	1.3	2.0	0.6
	PrecStd	7.2	2.3	1.7	1.4	1.3	1.0	1.5	2.1	1.9	2.2	2.2	1.5	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.8	46.8	58.3	69.1	79.2	82.3	85.7	80.4	73.4	63.0	53.8	49.1
		MCWB	42.7	42.8	46.8	54.2	60.7	63.3	67.2	66.1	62.5	56.7	52.4	48.0
	2%	DB	42.1	43.4	51.4	64.4	74.4	78.6	81.0	76.4	67.8	57.9	51.1	47.0
		MCWB	41.2	41.8	44.3	50.6	59.2	63.1	65.9	65.1	59.6	55.2	49.8	46.1
	5%	DB	40.6	41.4	46.6	59.7	70.1	74.0	77.5	72.9	64.6	56.0	49.4	45.2
		MCWB	39.7	40.1	42.3	48.7	57.2	61.2	64.6	63.5	58.7	54.0	48.2	44.4
	10%	DB	38.9	39.6	43.5	55.2	65.0	69.1	73.3	69.3	61.9	54.4	47.6	43.6
		MCWB	38.1	38.4	40.5	46.7	55.2	58.7	63.1	62.2	57.2	52.5	46.5	42.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.4	43.9	47.7	55.8	64.7	67.2	70.5	69.1	64.7	57.9	52.7	48.4
		MCDB	43.7	44.7	54.0	67.3	74.7	77.6	79.5	75.4	68.9	60.1	53.5	49.0
	2%	WB	41.4	42.1	45.1	52.6	61.7	64.5	68.0	66.5	61.5	56.0	50.1	46.4
		MCDB	41.8	42.8	49.5	60.6	69.8	75.0	76.7	72.8	66.0	57.6	50.8	47.0
	5%	WB	39.9	40.1	43.0	50.0	58.8	62.3	66.1	64.8	59.4	54.4	48.4	44.5
		MCDB	40.4	41.0	45.9	57.4	67.0	71.1	74.5	70.6	63.1	55.7	49.3	45.1
	10%	WB	38.3	38.4	41.1	47.4	56.0	60.2	64.2	63.0	57.9	52.9	46.6	42.9
		MCDB	38.9	39.4	43.3	53.3	63.2	67.5	71.5	67.8	61.1	54.2	47.5	43.5
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.9	14.2	19.7	20.8	20.0	19.2	17.5	16.4	13.0	8.8	8.1
		MCDBR	7.8	9.5	19.4	29.0	28.8	29.9	28.2	24.1	21.0	13.7	8.8	7.2
		MCWBR	7.9	8.0	13.3	17.2	15.2	16.0	14.6	13.6	13.7	10.7	8.3	7.2
	5% WB	MCDBR	7.8	8.4	16.8	24.8	24.4	24.9	23.6	19.4	18.2	11.8	8.1	7.3
		MCWBR	8.1	7.4	12.2	15.8	14.1	14.8	13.8	12.1	12.9	9.9	7.8	7.3
Clear-Sky Solar Irradiance	taub	0.300	0.327	0.345	0.381	0.378	0.369	0.391	0.391	0.359	0.346	0.322	0.289	
	taud	2.268	2.314	2.346	2.304	2.378	2.428	2.381	2.386	2.460	2.406	2.320	2.223	
	Ebn at Noon	181	228	257	263	272	277	267	258	251	219	171	149	
	Edh at Noon	16	23	29	35	35	34	35	32	26	21	15	13	
All-Sky Solar Radiation	RadAvg	123	316	756	1237	1619	1729	1593	1267	885	416	155	85	
	RadStd	24	50	76	170	154	133	203	121	108	67	29	10	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (69 neighbors)	+0.81	N/A	N/A	N/A	N/A	N/A	N/A	-227	-320	N/A	N/A

Nomenclature: See separate page